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Conventional versus Explainable Artificial Intelligence-Driven Threat Analysis in SIEM Systems: A Comparative Study

Atul Kumar Chauhan
Deptt. of CSE.
MITS-DU
Gwalior, India
atul@mitsgwalior.in

R.S. Jadon
Deptt. of CSE.
MITS-DU
Gwalior, India
rsjadon@mitsgwalior.in

Mir Shahnawaz Ahmad
Centre of Artificial Intelligence
MITS-DU
Gwalior, India
mirshahnawaz888@mitsgwalior.in

***Abstract*—Security Information and event management (SIEM) systems have evolved from rule-based threat detection to the implementation of advanced machine learning techniques. The continued development of Explainable Artificial Intelligence (XAI) enhances transparency and clarity in the complicated processes of identifying the threat, analysis, despite conventional SIEM systems being optimized for detection accuracy and processing efficiency. In the context of SIEM systems, this paper offers a thorough comparison of explainable AI-driven approaches with traditional approaches. The study breaks down the methodological differences, assesses performance indicators, and identifies the advantages, limitations, and issues of the application of both approaches and, therefore, reveals the research gap, namely the lack of standardized measures to evaluate the explanation, adversarial robustness, and obstacles in operational integration. The paper concludes by highlighting some future research directions that emphasize the importance of human-AI synergies, explainable models, and enhanced XAI methods in enhancing cybersecurity activities.**

***Index Terms*—Explainable AI, SIEM, Threat Analysis, Cybersecurity, Machine Learning, Security Operations Center**

I. Introduction

Security Information and event management (SIEM) systems are the backbone of contemporary cybersecurity operations because they take all sources of security events and aggregate them to investigate, respond, and detect threats. [1]. SIEM platforms combine Security Information Management (SIM) and Security Event Management (SEM) to provide centralized dashboards that improve threat visibility, improve incident response, and provide the necessary regulatory compliance. [2].

Traditional SIEM systems face critical challenges to ensure that they keep up-to-date with the increasing volume, velocity, and sophistication of cyber threats despite their critical role. These problems are limited scalability, rule-based detection, problems related to data integration, and significant reliance on manual validation. [3]. The history of SIEM has therefore deviated from the use of non-dynamic and signature-driven detection to the use of Artificial Intelligence (AI) driven methodologies that can identify more complex attack patterns and subtle variations or differences. [4].

The basic aim of traditional SIEM systems is to identify the threats and protect the organization data, administrators also check the alerts to protect, identify, and mitigate threats; these are the most important aspects in the traditional technique which was mostly dependent on signature-based threat models and data sets. Threat alerts validate, interpret, and respond to automated threat assessments by analysts, the inherent obscurity of these ”black box” models presents a significant challenge. [5] The occurrence of alert fatigue, which is characterized by a widespread lack of trust among analysts in the reliability of these alerts and an overwhelming tolerance to several security notifications, is frequently caused by the complexities of the data being presented. In addition, the basic difficulties associated with performing in forensic investigations and executing effective incident response protocols are a contributing factor. [6]. Explainable AI-based approach greatly enhances forensic traceability, which helps to comply with regulatory requirements and at the same time increases analysts' confidence in today's increasingly complex and multifaceted security procedures. [7] and XAI will play a very crucial role in cyber security to study and understand attacks features and also provide a very important contribution to the decision making of administrators. [8] Traditional rule-based SIEMs many of the time give the wrong alerts in high traffic environments, lack of contextual situational awareness, and are ineffective in detecting attacks, which can result in false positives and analyst burnout time to finding [9]. Thus, incorporating AI and, more recently, XAI into SIEM frameworks has become a promising innovation to increase the system's transparency, detection accuracy and reduce the human load [10]. Although XAI techniques like LIME and SHAP offer interpretable and actionable insights that enable analysts to comprehend and trust automated system decisions [11], AI-powered SIEM solutions leverage intelligence to uncover hidden patterns, anomalies, and novel threat vectors [12].

The SIEM system works like an investigator, it continuously monitors each incident and compares the incident with the

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intrusion signature dataset following the rules and regulations and takes action accordingly, but they are not able to identify new types of threat; here there is a need to identify the characteristics of the threat in which XAI technology is very helpful in identifying the threat and thus creating new rules for the future thereby reducing the work through false alerts of the administrators. That is why I try to give a comparative analysis of conventional SIEM system and explainable AI based methodologies for threat assessment.

A. Background

SIEM platforms have gradually changed from basic log-aggregation tools to sophisticated and complicated AI-enabled systems [13]. Traditional SIEMs were highly dependent on deterministic rule sets, signature matching, and statistical anomaly detection to identify known threats [14]. However, the exponential growth in network data, the dynamic nature of attack vectors, and the increase in false-positive rates reveal various limitations of such static and manually intensive and inefficient approaches [1].

The introduction of XAI has begun to reshape SIEM research and practice by offering human-centered interpretability and transparency in automated cybersecurity systems [15]. This new paradigm is critical to improve analyst decision confidence, ensure regulatory accountability, and bridge the trust gap between humans and AI-driven security systems [16].

B. Paper Organization

The remainder of this paper is organized as follows : Section Section - II expands with the related work. Section-III outlines the research methodology, objectives and significance. Section - IV presents the comparative analysis of methodological comparison, technical architecture, Operational performance, and implementation architectures. Section- V discusses the advantages, limitations, and trade-offs of both approaches. Section- VI presents summary of key findings and conclusions.

II. Related Work

Threat detection capabilities have been significantly enhanced by the combination of AI and machine learning with SIEM and intrusion detection systems [17]. Traditional SIEM systems frequently rely on set rules and signatures, which limits their ability to identify emerging or changing threats. [18]. Recent research has revealed that AI-powered SIEM and network intrusion detection systems (NIDS) outperform conventional techniques in terms of accuracy and scalability, particularly when working with large and complex datasets [19], [20].

However, there are significant challenges with regard to the interpretability and dependability of automated decisions when deep learning and complex models are used in cybersecurity. [21]. XAI techniques such as SHAP, LIME, DeepLIFT, Integrated Gradients, and others are frequently employed to address these problems and provide actionable insights for AI-generated alerts [22]–[24]. Their post-hoc explanation techniques enhance feature attribution, expedite incident response, and less alert fatigue of security analysts [25].

The successful use of SHAP and LIME in identifying critical features and clarifying detection results on large intrusion datasets was demonstrated by studies such as those conducted by Yang et al. [23] and Singh et al. [20]. Their frameworks demonstrated the critical importance of transparency for operational dependability and produced an accuracy of 96% While further work is required to reduce computational overhead and enhance real-time applications, comparative studies demonstrate that hybrid strategies that combine interpretable models with high-performance algorithms can obtain a harmonious blend of efficiency and explainability [23], [25].

Therefore, it can be said that these developments underscore how important it is to incorporate XAI into SIEM and NIDS architectures to improve operational efficiency, transparency, and trust in security initiatives.

III. Methodology

The paper uses a methodological approach of systematic literature review to thoroughly examine the present body of knowledge on the traditional versus explainable AI-based threat analysis in SIEM systems. We conduct our research in line with the systematic review guidelines in computer science and cybersecurity research works. [1], [22].

This research paper utilizes systematic literature review as a research method to provide a comprehensive overview of the current situation in the field of traditional and explainable AI-based threat analysis in SIEM systems. We conduct systematic reviews in line with the best practice of systematic reviews in computer science and cybersecurity research. The following are the key research objectives of this work:

Comprehensive Comparison: Detection algorithms, feature engineering, model architectures Systematically assess methodologies, architectures, and key performance characteristics of traditional, as opposed to XAI-based SIEM systems [8], [15], [20].

Performance Analysis: compare computational efficiency, and operational metrics across both approaches on various datasets and benchmarks [18], [26].

Implementation Assessment: Analyze the deployment challenges, integration requirements, and scalability considerations for XAI within operational SIEM environments [11], [19].

IV. Traditional and Explainable AI Approaches for Threat Analysis

This section seeks to compare and contrast traditional SIM methods with XAI methods with the goal to determine which technique is more suited to SIEM systems and produces better results.

In this paper, we compare and contrast explainable artificial intelligence (XAI) methods and technology with more conventional approaches, focusing on a number of key metrics for the efficient identification and thorough evaluation of

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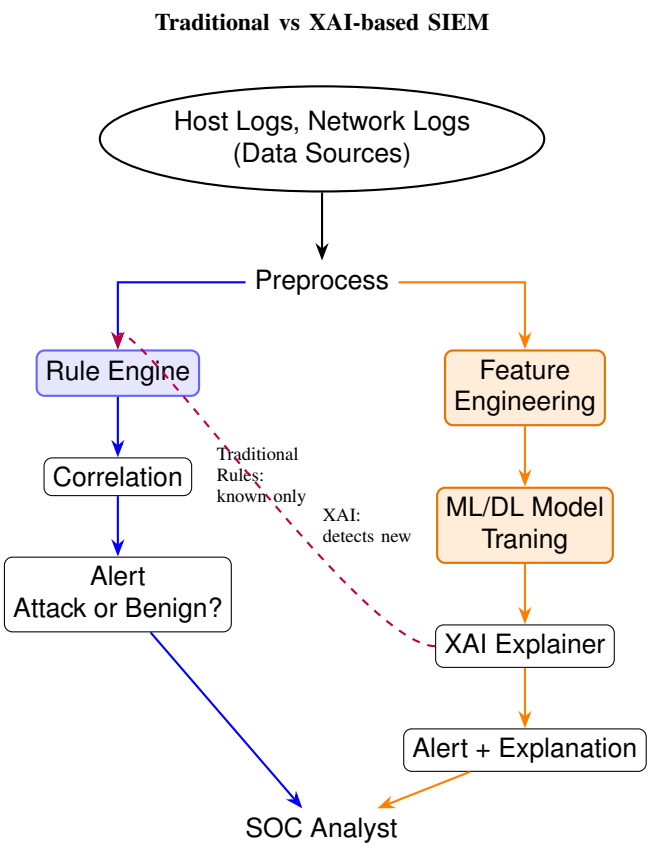

Fig. 1: Methodological comparison: Hybrid model and pipeline to find the explanation of attack

possible vulnerabilities that could manifest within the intricate networked environment's architecture.

1) Methodological Analysis: A thorough comparison between traditional and XAI-based threat detection and analysis pipelines within SIEM systems is provided by the methodological framework shown in Figure 1, which at first also shows the workflow of both approaches.

To identify potential threats system get input (Data Sources) integrates diverse streams such as logs from various devices, network log data from different sources, intrusion detection system alerts, and external threat intelligence data, which are used as the raw material for subsequent security analytics [2], [27]. In the initial period of community data processing, some very important activities, such as data parsing, normalization, and enrichment, were performed to ensure that the various data formats fit together for easier analysis later. That is, the preliminary processing layer, the architecture separates into two different but related routes.

Traditionally used static correlation rules based on signature base activity and signature databases to detect known threats [28]. Although this approach give the high interpretability and ease to validation for threats, it is inherently limited to well known attacks patterns and suffers from high false positive rates, placing a significant load on the Security Operations Center (SOC) analysts [18].

TABLE I: Technical Architectural Comparison between Traditional and XAI-Based SIEM

Aspect	Traditional SIEM	XAI-Based SIEM	References
Model Selection	Performance-optimized ensemble methods and deep learning	Interpretability–performance balance with post-hoc explainers	[6][7]
Feature Engineering	Discriminative power focus for maximum accuracy	Interpretable feature design with attribution methods	[1][2]
Training Objective	Minimize prediction error using standard ML metrics	Multi-objective: accuracy + explainability with SHAP/LIME integration	[5][6][7]
Output Format	Classification scores/labels without explanations	Predictions + explanations + confidence with natural language generation	[2][8]
Validation Metrics	Accuracy, F1, AUC for detection performance	Detection metrics + explanation quality (fidelity, stability)	[4][6][7]

On the other hand, the XAI pipeline engages in sophisticated feature engineering techniques that entitle statistical, temporal, and contextual attributes, feeding into advanced machine learning or deep learning models [29]. The XAI pipeline is enhanced by explainability modules, including various techniques such as SHAP and LIME, for transparency and the understanding about threat alerts, and the reason behind the generated alerts to take decisions accordingly [11], [22].

The hybrid doted arrow illustrated in the Figure 1 denotes potential synergies through which explain-ability facilitates the creation of innovative, human-comprehensible correlation rules that could be assimilated into traditional systems, thereby enhancing a human-in-the-loop framework. [25].

This review is an effort to direct future studies in the direction of creating hybrid models and XAI solutions that work in real-time that balance high performance with the necessary transparency, thus building more resilient and trustworthy cybersecurity frameworks.

2) Technical Architecture Analysis: The conventional methods and the XAI oriented method used for intrusion detection are analyzed on various aspects. Table I shows a tabular assessment of traditional and XAI-oriented SIEM designs. Conventional SIEM systems focus primary on performance enhancement through the utilization of ensemble or machine learning and deep learning frameworks that aim to optimize detection precision. However, these algorithms usually behave as somewhat confusing architectures, providing minimal or no understanding of their forecasting outputs.

On the other hand, XAI-based SIEM systems give the explanation for alert with the help of various tools(e.g., SHAP, LIME). These systems improve analysts' understanding of why specific alerts are triggered by associating features and contributing factors, and this feature reduces the time to take decisions at the time of attacks.

In traditional methods, they focus on statistical or domain specific threats, and detection mechanism, whereas XAI-based designs emphasize interpretable features that can be justified

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decisions, adhere to regulations, enhance understanding of AI reasoning, and visualized and explained at both local and global levels. [1]

Additionally, the representation of output progresses from basic labels or classification scores to enhanced predictions with explanations and confidence metrics, possibly even accompanied by narratives in natural language. Lastly, XAI-based SIEMs broaden the evaluation criteria to incorporate metrics for explainability fidelity, stability, and user trust, in addition to the traditional metrics of accuracy, precision, recall, and F1-score.

Overarchingly, this comparison emphasizes how explainable intelligence enhances analyst confidence, privacy intrusion detection, decreases false positives, and enables risk-informed decision-making in contemporary SOC settings.

3) Explanation Techniques: Statistical methods, rule engines, and classical machine learning models like Decision Tree, SVM, Naive Baye's, regression and Random Forest are commonly used in traditional intrusion detection systems. In these systems, explanations are usually derived from either hard-coded detection rules or feature importance metrics that are obtained after model training. In order to determine which variables have the greatest impact on the results, Random Forest models often use measures such as Gini importance or mean decrease in impurity to categorize features [6], [30]. This method gives an overarching impression of which features are important, but it doesn't explain why certain alerts or false positives occurred, so it's not very transparent when it comes to individual predictions. Security analysts must therefore deduce model reasoning either by inspecting hand-crafted thresholds or by correlating global feature rankings with historical alert patterns, often leaving the decision-making process a "black box" for individual events.

The SHAP summary plot in Figure 2 shows the impact of each feature on the finding explainable AI model's findings for intrusion detection, on the NSL-KDD dataset [31] using XGBoost. in particular, features most influence the model when predicting the type of attack, such as shown in Figure 2, high `src_bytes`, high `dst_bytes`, repeated request (`count`), errors etc. these high feature values give the direction toward the class of attack. That suggests a significant impact on the model's decision-making process. Recent studies have shown that the skewed or disproportionate importance of these features can directly lead to the creation of false alarms highlighted in [7], [30], [32].This may be especially noticeable if the model associates statistically extreme but benign values with attack behavior. By visualizing SHAP value distributions, analysts can determine which trait combinations are most frequently linked to inaccurate results. To reduce this effect, detection rules, thresholds, or feature engineering strategies can be modified [6], [33]. As a result, adding SHAP-based interpretability to SIEM systems will reduce false alert rates, enhance transparency, and make error analysis more practical.

The table-II presents a comprehensive details of various frameworks/approaches which use XAI for threat detection

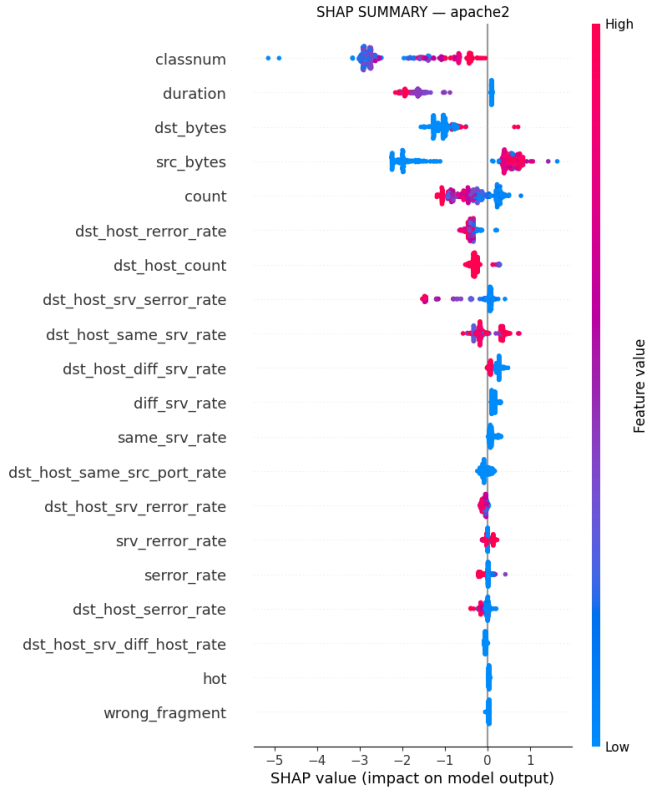

Fig. 2: SHAP Summary using XGBoost on NSL-KDD Dataset

and analysis on different benchmark network datasets. The key findings from the comparative analysis conclude that XAI methods offer significant benefits for result interpretation and operational auditability.

V. Discussion

Traditional techniques are very predicted and reliable, but they can obscure the clarity of security control systems and hub operations. However, the combination of ML/DL and explainable AI techniques not only maintains a high level of detection effectiveness but also clarifies feature-level transparency through the analysis of SHAP values. Security analysts can use the SHAP summary plots shown in (Figure effig:shap) to prioritize alerts and coordinate incident responses within the SOC by emphasizing the input features that have the biggest influence on model decisions.

The two main problems with traditional SIEM workflows that explainable AI solves head-on are the reliance on automated alerts by analysts and the lack of transparency and compliance with regulatory mandates regarding open decision-making processes [23]. The case for implementing XAI-enhanced SIEM frameworks is strengthened by the demonstrated capacity to evaluate model performance and shed light on specific predictions. The flawless combination of state-of-the-art machine learning and perceptive AI promotes the establishment of adaptable, clear, and durable SIEM systems, ideally attuned to the continuously changing environment of cybersecurity threat management.

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TABLE II: Comparative Analysis of Explainable AI Techniques for Threat Analysis and Intrusion Detection in SIEM

Reference	Dataset(s)	Model / Technique	XAI Method(s)	Key Findings / Contributions
Keshk et al. (2023) [29]	NSL-KDD, UNSW-NB15, TON_IoT	LSTM-based Deep Learning IDS	SHAP, Permutation Feature Importance (PFI), ICE, PDP	Explainable DL model reduced false positives and improved analyst confidence.
Moustafa et al. (2023) [34]	–	XAI for IoT intrusion detection frameworks	Multiple (SHAP, LIME, Anchors)	Highlighted open challenges and future research for XAI in operational SIEMs.
Sarker et al. (2024) [30]	–	Taxonomy of XAI for cybersecurity automation	General (SHAP, LIME, PDP, Grad-CAM)	Comprehensive survey linking AI trust, interpretability, and SIEM automation.
Gaspar et al. (2024) [35]	CICIDS2017, UNSW-NB15	MLP Neural Network for IDS	LIME, SHAP	Compared local vs global explanations; showed SHAP provided clearer insights for analysts.
Sharma et al. (2024) [31]	NSL-KDD, UNSW-NB15	Deep Learning-based IDS	SHAP, LIME	Demonstrated interpretable deep models outperforming traditional IDS with improved transparency.
Nugraha et al. (2025) [36]	UNSW-NB15	XGBoost with hybrid feature selection	SHAP, LIME	Proposed two-stage framework combining feature reduction and explainable learning for lightweight IDS.
Adaptable IDS (2025) [37]	IoT Network Traffic	Incremental Learning Neural Models	Generic explainers (LIME, SHAP)	Introduced adaptive and evolving explainable IDS improving concept drift handling.
Mohale & Obagbuwa (2025) [38]	UNSW-NB15	Multiple ML models (RF, SVM, KNN)	SHAP, LIME, ELI5	Compared several classifiers with interpretability scores; evaluated transparency vs accuracy trade-offs.
Uccello et al. (2025) [39]	Custom SIEM Dataset	Hybrid SIEM rules + ML framework	xAI-derived rules (SHAP-driven)	Integrated explainable AI with rule-based SIEM to enhance DDoS and anomaly detection.
Almheiri et al. (2025) [40]	Simulated enterprise network	Hybrid DL + statistical modeling	XAI for sustainable cybersecurity	Emphasized model transparency and energy-efficient explainable detection systems.

Nonetheless, various constraints deserve thorough examination. The elaborate computational necessities of XAI methodologies may escalate delays and resource consumption, thereby possibly hindering scalability within large corporate frameworks [25].

VI. Conclusion

The aim of this research paper is to identify the basic difference between traditional methods and XAI for thread detection. in this regard, we compare the traditional methods with XAI methods in threat detection within the SIEM application with the help of literature review.

As the future of security operations become increasingly complex, the introduction of explainable artificial intelligence for security operations is not just a technological change, but an overhaul of security operations to a more transparent, responsible, and efficient cybersecurity operations. The evidence studied in this paper gave a road-map on how to operate through this change as well as making the most possible out of it and reducing risks.

The literature evidence indicated that those organizations that invest in XAI capabilities in the present and tackle the challenges identified in the paper, will be in a better speed to manage the dynamic threat landscape of the future. This long-term cooperation of researchers, practitioners, and technology providers will be necessary to change the situation regarding technical, organizational, and ethical issues outlined in this detailed review.

It seems that the path toward XAI-empowered SIEM systems is unavoidable, and it was predetermined due to the growing number and complexity of cyber attacks that overwhelm the rule-based systems. Nonetheless, to achieve the potential of Explainable AI in cybersecurity, the present issues of the quality of explanations, the complexity of integration, and approaches to evaluation should be resolved.

The dynamics of the future must be directed to make XAI more ubiquitous within the SIEM framework in order to examine both the scalability and latency aspects of XAI methods within the context of real world, high traffic SIEM data. Moreover, the evaluation of diverse state-of-the-art ex-

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plainability procedures and adaptive learning policies will be a path device on the way to developing intelligent SIEM of the next generation.

In the coming years, it is probable that fully developed XAI systems with standardized interfaces, better explanation methods and tested patterns of deployment will emerge.

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